Tracing the Flow: Modeling Contamination Spread in a Water Distribution Network

NGSS-Aligned Lesson Plan | Quinn MacKenzie, AI-STEM Coordinator, Timothy Christian Schools

Grade Band	Subject Tie-In	Duration	Group Size
9–12 (Physical Science / Engineering / AI-STEM)	Environmental engineering, systems modeling, computer science	One 50-min period	2 groups of 6–12 students

NGSS Standards Addressed

Performance Expectations	
HS-ETS1-2 — Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems. HS-ETS1-3 — Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs. HS-ESS3-4 — Evaluate or refine a technological solution that reduces impacts of human activities on natural systems.	**Science & Engineering Practices** Developing and Using Models; Analyzing and Interpreting Data; Constructing Explanations and Designing Solutions **Crosscutting Concepts** Systems and System Models; Stability and Change **Disciplinary Core Ideas** ETS1.B: Developing Possible Solutions; ETS1.C: Optimizing the Design Solution

Learning Objectives

By the end of this lesson, students will be able to:

- Model how a substance introduced at one point in a piped network reaches the rest of the system over time, using a physical simulation.
- Explain, using evidence from both the physical model and a computer simulation, why an entry point's position in a network — not just the substance itself — determines how far and how fast contamination spreads.
- Interpret and compare data (arrival times, concentration) generated by a hands-on activity and by a Python-based EPANET/WNTR simulation of the same network.
- Evaluate trade-offs in proposed engineering responses (e.g., sensor placement, valve shut-off) using criteria such as detection speed and network coverage.

Materials

- Floor tape or cones to lay out a network of 6–7 “junction” stations (mirrors the R1–J6 network used in the research simulation)
- Color-coded note cards: blue = water, red = contaminant
- Stopwatch or timer, whiteboard or poster paper to log arrival times per round
- Laptop/projector with the class WNTR/EPANET Python simulation for the demonstration network

Lesson Sequence: 5E Lesson plan

Phase	Time	Activity
Engage	**5 min**	Pose the question: “If something contaminated our school’s water at one fountain, how would we know how far it spread — and how fast?” Brief class discussion.
Explore	**25 min**	Students stand at taped “junction” stations arranged in the same network layout used in the research model. Each round represents one simulated hour: students holding water cards pass them to connected neighbors; a contaminant card enters at a student-chosen junction and moves along with the water cards from that point on. The class logs which junction the contaminant reaches each round on a shared network diagram.

Explain	**15 min**	Teacher runs the same scenario in the Python/WNTR simulation and displays the hour-by-hour concentration table and network snapshots. Class compares the simulation's arrival times to their physical model and discusses why they match (or don't).
Elaborate	**15 min**	In groups, students pick a different starting junction, predict how far and how fast contamination will spread, then check their prediction against the simulation's automated vulnerability scan, which ranks every junction by reach and speed. Groups discuss where a utility should prioritize sensors.
Evaluate	**10 min**	Exit ticket: students explain, in writing, why the same contamination event produces different outcomes depending on where it enters the network, citing evidence from both the physical activity and the simulation data.

Assessment

- Formative: teacher observation of group reasoning during the Explore and Elaborate phases; accuracy of logged arrival times.
- Summative: exit-ticket written response, scored against a short rubric for correct use of evidence, systems-thinking vocabulary, and connection between the physical and computational models.

Differentiation & Extensions

- Support: provide a pre-drawn network diagram and simplified round-by-round tracking sheet for students who need a visual scaffold.
- Extension: students with prior programming experience modify the source node and timing parameters directly in the Python script and predict the output before running it.
- Cross-curricular tie-in: connect to environmental science units on public health and infrastructure, or to AP Computer Science units on modeling and simulation; Share this lesson with those respective teachers.

Key Vocabulary

Node / Junction	A connection point in the network (e.g., a pipe intersection or fixture) where water can be drawn or contamination can enter.
Contaminant source	The location and strength of a substance introduced into the network, modeled as a rate of mass entering the system over time.
Concentration	The amount of contaminant present in the water at a given node, typically measured in milligrams per liter (mg/L).
Systems model	A simplified representation of a real system (here, a water network) used to predict how it behaves under different conditions.
Vulnerability	A measure of how much of a network is affected, and how quickly, when contamination enters at a given point.

Connection to Ongoing Research

This lesson is directly adapted from an active research and curriculum-development project modeling contamination spread in water distribution networks using WNTR (Water Network Tool for Resilience) and the EPANET simulation engine in Python. The classroom network layout, the concept of a configurable contaminant source, and the vulnerability-ranking activity all mirror methods used in that research — giving students a physical, tactile entry point into the same systems-thinking and computational-modeling skills used in real infrastructure engineering and public health preparedness work.